UCR University College Roosevelt

Fall 2026 cohort

Astronomy at University of Groningen

Starting interests: **Space exploration and astronomy**

Official programme ↗

Astronomy and astrophysics

Introduction Astronomy · 5 EC
Physics of Stars · 5 EC
Physics of Galaxies · 5 EC
Radiative Processes in Astrophysics · 5 EC
Advanced Electrodynamics · 5 EC
Astrophysical Hydrodynamics · 5 EC
Interstellar Medium · 5 EC

Physics and mathematics

Mechanics and Relativity · 10 EC
Electricity and Magnetism · 10 EC
Thermal Physics · 10 EC
Quantum Physics 1 · 5 EC
Quantum Physics 2 · 5 EC
Waves and Optics · 5 EC
Calculus 1 (for Physics) · 5 EC
Calculus 2 (for Physics) · 5 EC
Linear Algebra (for Physics) · 5 EC
Mathematical Physics · 5 EC
Complex Analysis · 5 EC

Computing, data and observation

Introduction to Programming and Computational Methods · 5 EC
Numerical Methods · 5 EC
Statistics for Astronomy · 5 EC
Observational Astronomy · 5 EC

Exploration, instruments and responsible technology

Minor: Instrumentation & Informatics · 30 EC

Earth, planets and life

Introduction to Planetary Science · 5 EC

Research and academic development

Physics Lab: Skills · 5 EC
Astronomy Bachelor Research Project · 15 EC

UCR does not currently offer astronomy or physics courses. The UCR programmes shown here therefore build around the closest feasible combinations of mathematics, computing, observation, Earth systems, life science and technology.

Selected minor route: Instrumentation & Informatics (30 EC).

UCR University College Roosevelt

Fall 2026 cohort

Computing, Data and Earth Observation

Starting interests: **Space exploration and astronomy**

UCR courses ↗

Physics and mathematics
Advanced Calculus · 7.5 EC
Linear Algebra · 7.5 EC
Mathematical Methods · 7.5 EC
Logic & Discrete Mathematics · 7.5 EC
Linear Systems · 7.5 EC
Computing, data and observation
Introduction to Programming · 7.5 EC
Computational & Numerical Methods · 7.5 EC
Probability & Statistics · 7.5 EC
Earth Observation · 7.5 EC
Introduction to Data Science · 7.5 EC
Machine Learning · 7.5 EC
Image Processing & Computer Vision · 7.5 EC
Neural Networks & Deep Learning · 7.5 EC
Optimization Methods in AI · 7.5 EC
Reinforcement Learning · 7.5 EC
Algorithms & Data Structures · 7.5 EC
Networks & Operating Systems · 7.5 EC
Database Management Systems · 7.5 EC
Earth System Science Tools · 7.5 EC
Exploration, instruments and responsible technology
Robotics · 7.5 EC
Consumer Product Design · 7.5 EC
Earth, planets and life
Sustainability Essentials I · 7.5 EC
Environmental Biogeochemistry · 7.5 EC
Research and academic development
Personal & Professional Development · 7.5 EC

UCR does not currently offer astronomy or physics courses. The UCR programmes shown here therefore build around the closest feasible combinations of mathematics, computing, observation, Earth systems, life science and technology.

UCR does not offer astronomy or physics courses; this programme builds the closest feasible depth through mathematics, computing, data and observation.

UCR University College Roosevelt

Fall 2026 cohort

Earth Observation, Robotics and Sustainable Technology

Starting interests: **Space exploration and astronomy**

UCR courses ↗

Physics and mathematics

Linear Algebra · 7.5 EC

Mathematical Methods · 7.5 EC

Logic & Discrete Mathematics · 7.5 EC

Computing, data and observation

Introduction to Programming · 7.5 EC

Computational & Numerical Methods · 7.5 EC

Probability & Statistics · 7.5 EC

Earth Observation · 7.5 EC

Introduction to Data Science · 7.5 EC

Machine Learning · 7.5 EC

Image Processing & Computer Vision · 7.5 EC

Optimization Methods in AI · 7.5 EC

Earth System Science Tools · 7.5 EC

Exploration, instruments and responsible technology

Robotics · 7.5 EC

Consumer Product Design · 7.5 EC

Renewable System Design · 7.5 EC

AI Ethics · 7.5 EC

Earth, planets and life

Sustainability Essentials I · 7.5 EC

Sustainability Essentials II · 7.5 EC

Dynamic Earth · 7.5 EC

Environmental Biogeochemistry · 7.5 EC

Earth's Resources · 7.5 EC

Climate Change · 7.5 EC

Research and academic development

Research in a Sustainable Delta · 7.5 EC

Personal & Professional Development · 7.5 EC

UCR does not currently offer astronomy or physics courses. The UCR programmes shown here therefore build around the closest feasible combinations of mathematics, computing, observation, Earth systems, life science and technology.

UCR does not offer astronomy or physics courses; this programme combines quantitative tools with Earth science and exploration technology.

UCR University College Roosevelt

Fall 2026 cohort

Earth Systems, Technology and Life Sciences

Starting interests: **Space exploration and astronomy**

UCR courses ↗

Physics and mathematics
Mathematical Methods · 7.5 EC
Computing, data and observation
Introduction to Programming · 7.5 EC
Earth Observation · 7.5 EC
Introduction to Data Science · 7.5 EC
Machine Learning · 7.5 EC
Image Processing & Computer Vision · 7.5 EC
Neural Networks & Deep Learning · 7.5 EC
Earth System Science Tools · 7.5 EC
Exploration, instruments and responsible technology
Robotics · 7.5 EC
Consumer Product Design · 7.5 EC
Renewable System Design · 7.5 EC
AI Ethics · 7.5 EC
Earth, planets and life
Sustainability Essentials I · 7.5 EC
Sustainability Essentials II · 7.5 EC
Dynamic Earth · 7.5 EC
Environmental Biogeochemistry · 7.5 EC
Earth's Resources · 7.5 EC
Climate Change · 7.5 EC
Introduction to Biodiversity · 7.5 EC
The Science of Evolution · 7.5 EC
Foundation of Life, Health & Biomedicine · 7.5 EC
Biochemistry · 7.5 EC
Research and academic development
Research in a Sustainable Delta · 7.5 EC
Personal & Professional Development · 7.5 EC

UCR does not currently offer astronomy or physics courses. The UCR programmes shown here therefore build around the closest feasible combinations of mathematics, computing, observation, Earth systems, life science and technology.

UCR does not offer astronomy or physics courses; this programme broadens into Earth systems, life science, data and exploration technology.